

Corresponding Kinetic Energy in Relativity

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Abstract

In this paper, we illustrate a difficulty while combining relativity to statistical mechanics, hence review the definition of kinetic energy in relativity.

Difficulty

We know in classical, the momentum distribution of a particle having mass m is

$$dP = d^3\mathbf{p} \left(\frac{\beta}{2\pi m} \right)^{\frac{3}{2}} e^{-\beta K}$$

given temperature T . However, in relativity $v \leq c$, we have to rethink “momentum” and “kinetic energy” to avoid arbitrary $\mathbf{v}$. Majority gives

$$\mathbf{p} = \gamma m \mathbf{v}, K = (\gamma - 1)mc^2$$

but it involves special functions and change of variables seems to be forbidden, make us suspect if it's correct.

Review

We know the motivation of K part come from

$$dK = \frac{d(\gamma m \mathbf{v})}{dt} \cdot \mathbf{v} dt$$

but there is a caveat in choosing observer time or own time. If we choose own time:

$$dK = \frac{d(\gamma m \mathbf{v})}{d\tau} \cdot (\gamma \mathbf{v}) d\tau$$

it motivates $K = \frac{p^2}{2m}$, identical to the classical form. Expanding

$$\frac{1}{2}mv^2 \left(\frac{1}{1 - \left(\frac{v}{c}\right)^2} = 1 - \left(\frac{v}{c}\right)^2 + \left(\frac{v}{c}\right)^4 - \dots \right)$$

still reduce to $\frac{1}{2}mv^2$ in Newton limit. Hence we suggest

$$\mathbf{p} = \gamma m \mathbf{v}, K = \frac{p^2}{2m}$$

this avoids special functions and is compatible to relativity.

Summary

Although we find a caveat in choosing derivative lead to another possibility, no one over the other either in observation or theoretical. We're not sure which is correct but it makes us think how times play a role in thermodynamics.

Reference

- [1] Robert H. Swendsen “An Introduction to Statistical Mechanics and Thermodynamics”
- [2] David Tong “Dynamics and Relativity”